

Bright Coffee Sales Analysis

Project Overview

The Bright Coffee Sales Analysis is a data analytics project focused on analysing coffee shop transaction data to understand sales performance, customer purchasing patterns, product performance, and transaction trends.

The project transforms raw transaction data into a clean and structured dataset that can be used to generate meaningful business insights and interactive dashboards.

Project Objectives

The main objectives of the project are to:

- Analyse overall coffee shop sales performance.
- Identify the highest- and lowest-performing products.
- Analyse sales by product category and product type.
- Examine transaction volumes and quantities sold.
- Identify sales patterns by day, month, and time of day.
- Analyse transaction values and purchasing behaviour.
- Identify high- and low-value sales transactions.
- Create useful metrics and KPIs for business performance.
- Develop dashboards that communicate important sales insights.
- Provide data-driven recommendations for improving sales performance.

Data Cleaning and Transformation

The raw Bright Coffee dataset was cleaned and transformed using SQL in Databricks.

Key data preparation activities included:

- Handling missing and blank values.
- Standardising product categories and product types.
- Converting transaction quantities into appropriate numeric formats.
- Cleaning and converting unit prices.
- Converting transaction dates into usable date formats.
- Extracting day, month, year, and day-of-week information.
- Extracting transaction hours from time fields.
- Creating transaction time buckets.
- Calculating total transaction value.

- Creating sales value categories.
- Creating quantity categories.
- Preparing the final dataset for analysis and dashboard development.

Key Metrics

The analysis includes metrics such as:

- Total Sales
- Total Transactions
- Total Quantity Sold
- Average Transaction Value
- Average Unit Price
- Sales by Product
- Sales by Product Category
- Sales by Product Type
- Sales by Store Location
- Sales by Day
- Sales by Month
- Sales by Time of Day

Key Business Questions

The project investigates questions such as:

- Which products generate the highest sales?
- Which product categories perform best?
- Which products have lower sales performance?
- What days generate the highest transaction volumes?
- Which times of day have the highest sales activity?
- How does transaction quantity affect sales value?
- Which store locations generate the most sales?
- What are the patterns in customer purchasing behaviour?
- Which products or time periods provide opportunities for improving sales?

Tools and Technologies

- Databricks – Data cleaning, transformation, and SQL analysis.

- SQL – Data manipulation, aggregation, calculations, and analytical queries.
- Excel – Pivot tables, analysis, and dashboard development.
- Power BI / Looker Studio – Data visualisation and interactive dashboards.
- GitHub – Version control and project documentation.

Project Workflow

The project follows a complete data analytics workflow:

Raw Data → Data Cleaning → Data Transformation → Exploratory Analysis → KPI Development → Visualisation → Business Insights

This workflow demonstrates how raw transactional data can be converted into a structured analytical dataset and then presented through dashboards.

Skills Demonstrated

- SQL
- Databricks
- Data cleaning
- Data transformation
- Exploratory data analysis
- Data aggregation
- KPI development
- Sales analysis
- Data visualisation
- Dashboard development
- Business intelligence
- GitHub and version control

Project Outcome

The Bright Coffee Sales Analysis provides a structured view of coffee shop sales performance and customer transaction patterns. The resulting analysis and dashboards can help identify important sales trends, understand product performance, and highlight areas that may require further investigation or business attention.

This project forms part of my Data Analytics Portfolio and demonstrates practical experience in SQL, Databricks, data preparation, business analysis, and data visualisation.